

TVRTKO ŠAPINA

Third-year Mathematics student with a strong interest in quantitative finance, mathematical modelling, and machine learning. Experienced in using Python for data science, automation, and financial analysis projects.

EDUCATION

Bachelor's Degree in Mathematics | Faculty of Science, University of Zagreb

EXPECTED GRADUATION 2027

Relevant coursework: Probability, Statistics, Numerical mathematics, Linear algebra, Calculus, Differential equations, Measure theory, DSA

WORK EXPERIENCE

Student data scientist | Hrvatski Telekom

JUL 2026 | SEP 2026

Performed feature engineering for a customer credit risk assessment model and extracted relevant data from sql databases.

IC Academy | InterCapital

SEP 2026

Selected through an asset-management assignment organized by a leading independent investment company in Southeast Europe. The program offered a comprehensive introduction to InterCapital's business, including macroeconomic analysis, valuation of bonds, equities and real estate, as well as trading.

Internship in the Controlling and Risk Management Department | E.ON

MAR 2025 | JULY 2025

Developed a Python application for automating the extraction of market data from the HUDEX energy exchange.

Built a Python-based tool that uses polynomial interpolation to estimate monthly natural-gas prices from quarterly and annual forward prices.

Photography and digital marketing - freelance projects | Nu bar, Nec, Villa Madomi

2022 | 2025

PROJECTS

- **Option Pricing Toolkit**

Developed a Python toolkit for pricing European, American, and Bermudan call and put options using CRR binomial tree, Black-Scholes and Monte Carlo simulation. Implemented Euler-Maruyama, tamed Euler, and Milstein numerical schemes for solving stochastic differential equations. Available at: <https://optionpricingtoolkit-dewbzm3qzn6d2rh3caof8n.streamlit.app/>

- **Banknote Authentication**

Performed data preprocessing, explored clustering-based approaches, trained supervised learning models, and compared the performance of logistic regression, SVM, decision tree and gradient boosting. Available at: <https://github.com/tvrtkosapina/portfolio.git>

SKILLS

- **Technical:** Data analysis, MS Excel
- **Languages:** Croatian (native), English (C1), German (A1)
- **Programming languages:** Python (scikit-learn, Pandas, NumPy, SciPy), C, basic C++, basic SQL